



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2025-2026
COURSE HANDOUT

Date: 03.08.2026

In addition to part I (General Handout for all courses appended to the time table) this portion gives further specific details regarding the course.

Course No. : PHA G541
Course Title : Computer Aided Drug Design
Instructor-in-Charge : Prof. S. Murugesan
Instructor : As given in the time table

1. Course Description:

3D structure and function of bio-molecules; targets of drugs and design principles; molecular modeling methodologies; quantitative structure-activity relationships; chemical compound databases and search tools; interactive graphics in drug design; molecular surfaces and algorithm of automated docking of drugs into receptor sites; receptor mapping; introduction to molecular modeling and docking software.

2. Scope and Objectives:

To develop an appreciation of inter- and intra- molecular energies and forces. To introduce the techniques of computational chemistry and to demonstrate their application to the design of drugs. To demonstrate the use of simulation as an alternative to experimental work. On completion the student should be aware of the scope and limitations of the various theoretical and computational tools available to the pharmaceutical scientist and of the application of these techniques in the design of novel drugs. The student will also have gained insight into the practical implementation of the methods of molecular and bio-molecular modeling.

3. Text Book:

1. Manfred E Wolff- "Burger's Medicinal Chemistry and Drug Discovery", Wiley-Interscience, Seventh Edition, N.Y., 2010, Vol. I - VIII.

4. Reference Books:

1. P.K.Larson, T.Liljefors & U.Madsen- "Textbook of Drug Design and Discovery", 1st edition, Indian edition, Chennai, 2004.
2. Cramer, C. J. Essentials of Computational Chemistry: Theories and Models; Second Edition, Wiley, 2004.

Besides the above relevant information on the topics are also available in the following:

1. Annual Reports in Medicinal Chemistry - Academic Press Inc, Various volumes.
2. Journal of Medicinal Chemistry-ACS-Different Volumes.
3. Chemical Reviews-ACS-Various Volumes.
4. Journal of Molecular Modeling - Springer-Verlag.
5. Journal of Computer Aided Molecular Design – Kluwer Academic Publishers.
6. Drug Discovery Today – Elsevier Publishers.

5. Course Plan:

Lect No.	Learning Objectives	Topics to be covered	Reference	Learning Outcome
1-3	Introduction	Drug design overview, reason for failures of drugs, stages where computational methods can be used	T.B.	Information about various aspects of computer aided drug design approaches used in novel drug discovery process
4-6	Drug likeliness	Drug likeliness criteria, lead likeliness, hurdles in drug discovery	T.B.	Information about selection of drug like



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

				as well as lead like molecules
7-12	Protein structure determination and analysis	Crystallography, 2D NMR, homology modeling, de novo design and protein folding	T.B and Ref.(1)	Information about selection of proteins for drug design
13-16	Homology model building	Concept of homology and similarity, steps involved in homology model building	Reference (2)	Information about making of proteins for drug design
17-20	Compound library design	Targeted Vs diverse libraries, fragment Vs reactions approach, synthetic accessibility and compound selection techniques	T.B. and Ref.(1)	Information about creation of ligand database for drug design
21-23	QSAR	2D QSAR, descriptors, process for generating equation, 3D QSAR process, 3D QSAR packages	T.B.	Information about various aspects of 2D and 3D quantitative structure activity relationship studies
24-26	ADMET predictions	Oral bioavailability, plasma half-life, BBB permeability, toxicity predictions using computational methods	Various sources	Information about selection of NCE's with better pharmacokinetic and lesser toxic profile
27-29	Pharmacophore modeling	Components of pharmacophore tool, creating pharmacophore model from active compounds and active site, searching databases, reliability of results	T.B. and Ref.(1)	Information about generation of new pharmacophore and it's validation for drug design
30-33	Docking	Molecular mechanics, force fields, search algorithms, scoring, validation of results, comparison of existing docking programmes, preparation of target, design of ligands, analysis of results	T.B. and various databases	Information about finding out binding free energy of molecules upon interaction with target
34-36	Drug design process for known target	Finding out initial hits, compound refinement procedures, ADMET and drug resistance mechanisms	Ref. (1)	Information about various processes involved in the design for known target
37-40	Drug Design process for unknown and other targets	Initial hits, compound refinement and ADMET; DNA, RNA, receptors as targets, targets inside cells, CNS, etc	T.B and Ref.(1)	Information about various processes involved in the design for unknown target



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Remarks
Mid-sem Exam	90 min	30	06/10 FN2	CB
Continuous assessment including practicum component*		35		CB & OB (or) CB / OB
Comprehensive Exam	180 min	35	05/12 AN	CB & OB (or) CB / OB

*Continuous assessment will be based on theory covered in class. Components, number and weightage of continuous evaluation will be announced in the class. It may be consisting of take home assignment, tutorials, laboratory Day to Day work, presentation, seminars, Quiz, Viva-voce etc.

Closed Book Exam: No reference material of any kind will be permitted inside the exam hall.

Open Book Exam: Use of any printed / written reference material (books and notebooks) will be permitted inside the exam hall. Loose sheets of paper / Photocopied materials will not be permitted. Computers of any kind will not be allowed inside the exam hall. Use of calculators will be allowed in all exams. Exchange of any material will not be allowed.

Reading Assignments: Students are advised to read, collect additional information on the above mentioned topics from journals and other online sources.

7. Further Information:

Attendance: Attendance is compulsory as per the Institute policy, regularity in theory and practical classes will be decisive factor during make-up and grading, especially in borderline cases as well as make-up's.

Grading Procedure: It is not mandatory to award all the grades (A to E); subjective judgment will be exercised while awarding grades. In borderline cases subjective judgment will be exercised to decide the final grade. The student shall not be considered exposed to the course, unless he/she demonstrates appreciable skill in both laboratory and theory component of the course.

Make-up policy: Generally, make-up will be considered for regular students only and make-up's are not given as a routine. It is solely dependent on the "genuineness" of the circumstance under which a student fails to appear in a scheduled evaluation component. However, the make-up application should be personally given to Instructor-in-Charge and not slipped into the chamber of the Instructor-in-Charge.

Chamber Consultation Hour: To be announced in the class.

Notices: Concerning this course will be displayed on Pharmacy NB

Instructor-in-Charge
PHA G541